MANCHESTER 1824
The University of Manchester

AERO60492 – Autonomous Mobile Robots

Lecture 6 – Rotation representations and coordinate systems

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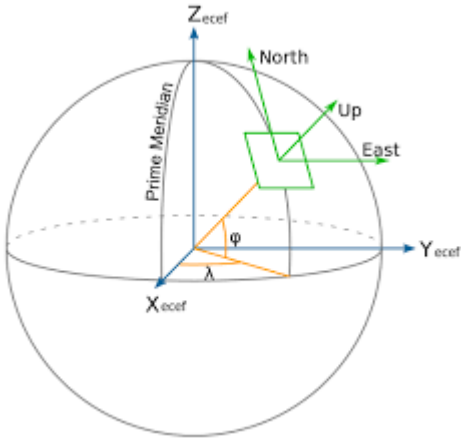
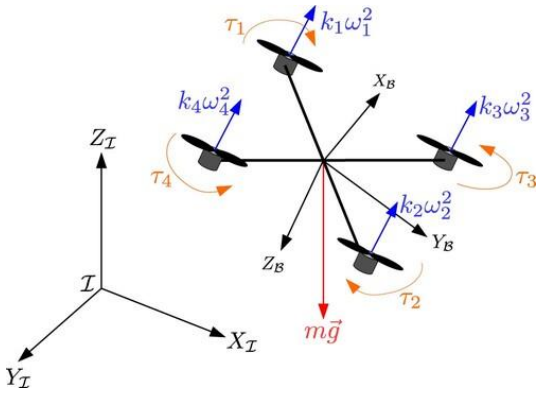
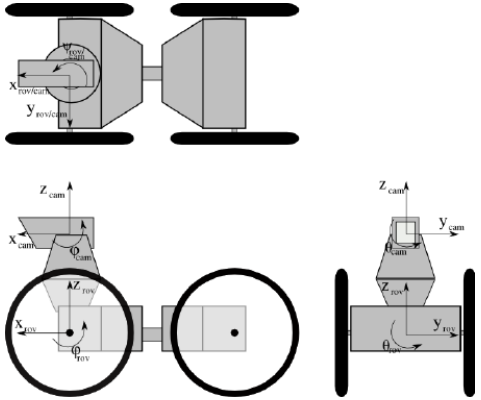
Aims

- Coordinate systems encountered for AMR
- General rules and equations for building general coordinate systems
- Rotations: representations, converting, and applying





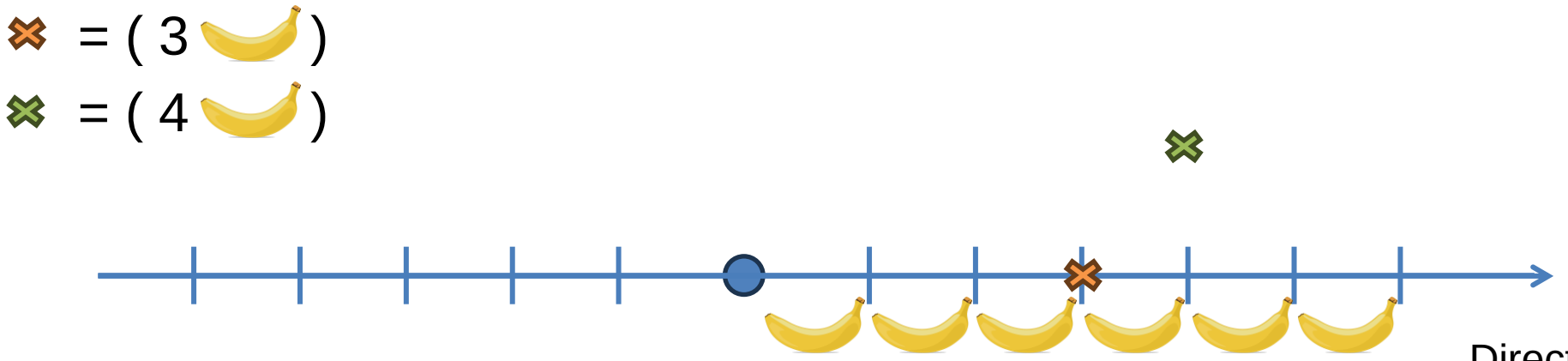


Frames of reference

- A abstract coordinate system, specified by,
  - origin
  - orientation/system
  - scale/units
- Also called coordinate frames
  - e.g. Chapter 2 “Small Unmanned Aircraft”, Beard & McLain
- Provides a method to know how multiple bodies are orientated and positioned relative to each other

Coordinate System

Linear 1D - Simplest coordinate is distance along a straight line from an origin



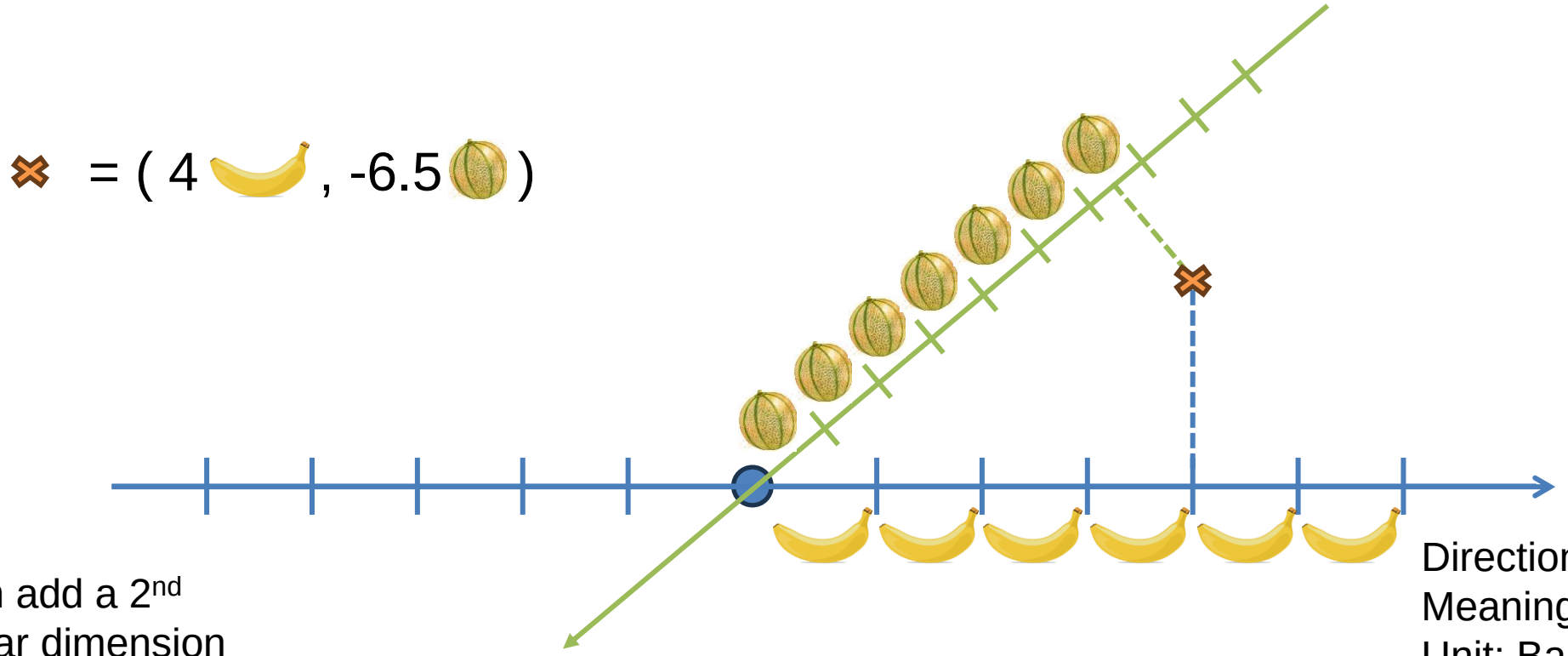
Direction: +ive
Meaning: Position
Unit: Banana

Origin of reference frame

A point (or body) within reference frame

Coordinate System

Linear 2D - Can add a 2nd dimension for 2D position



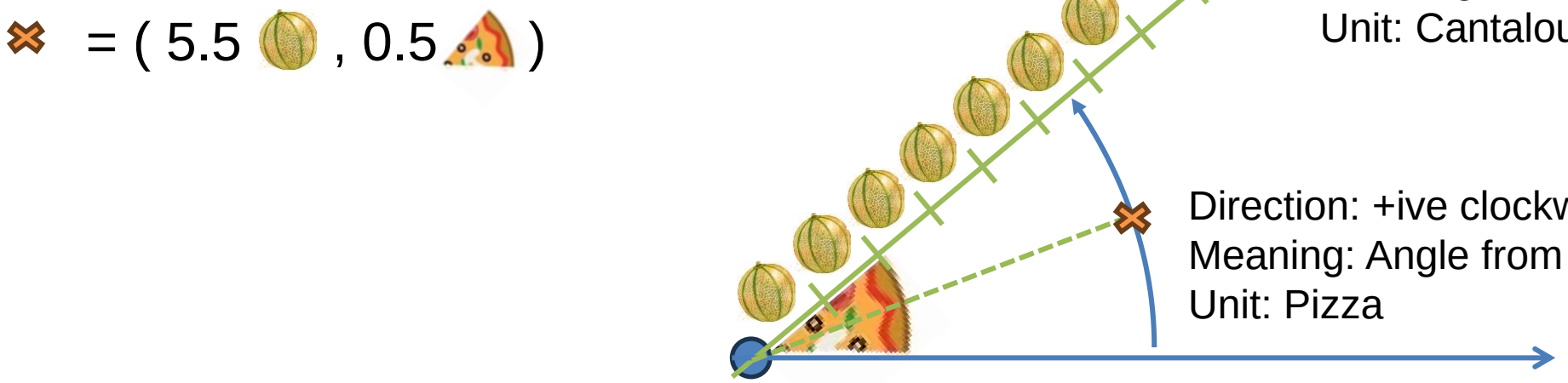
Can add a 2nd linear dimension

Direction: +ive
Meaning: Position (2nd dimension)
Unit: Cantaloup

Direction: +ive
Meaning: Position
Unit: Banana

Coordinate System

Non-linear 2D



Direction: +ive
Meaning: Distance from origin
Unit: Cantaloup

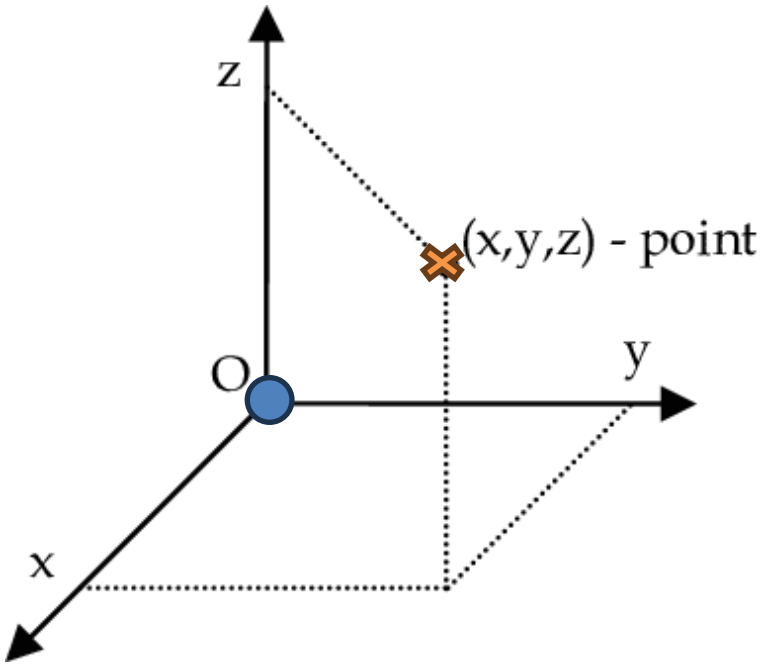
Direction: +ive clockwise
Meaning: Angle from datum
Unit: Pizza

Take away message: Coordinate systems can be anything,
Providing the origin, direction, and scale/units are defined.

Cartesian coordinates

Linear 3D

- 3 mutually orthogonal axes with identical scale
- Useful because linear algebra and calculus can be applied

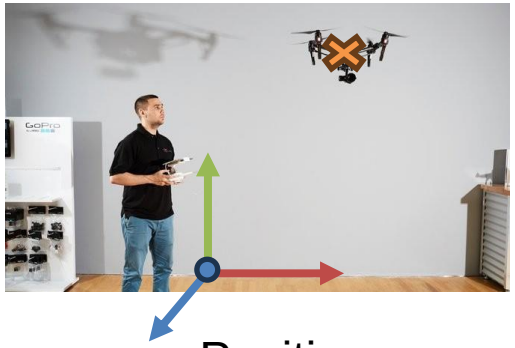


Coordinate systems

- Mostly Cartesian for Autonomous Mobile Robots
- Mostly 3-dimensional for Autonomous Mobile Robots
- Some exceptions
  - Spherical coordinates
  - Cylindrical coordinates
  - Image coordinates
- Multiple coordinate systems might be nested for modelling the behaviour of a single robot

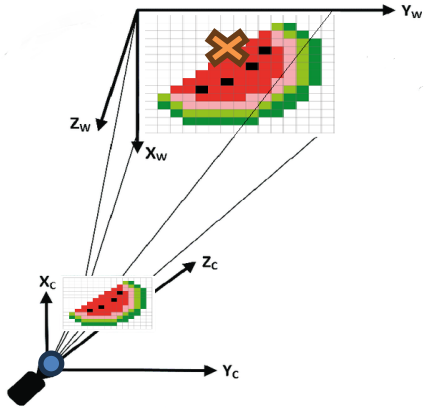
Nested frames of reference

Position relative to Earth

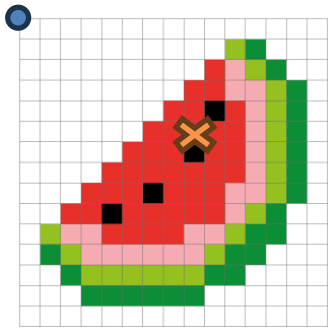


Position relative to room

Position relative to robot



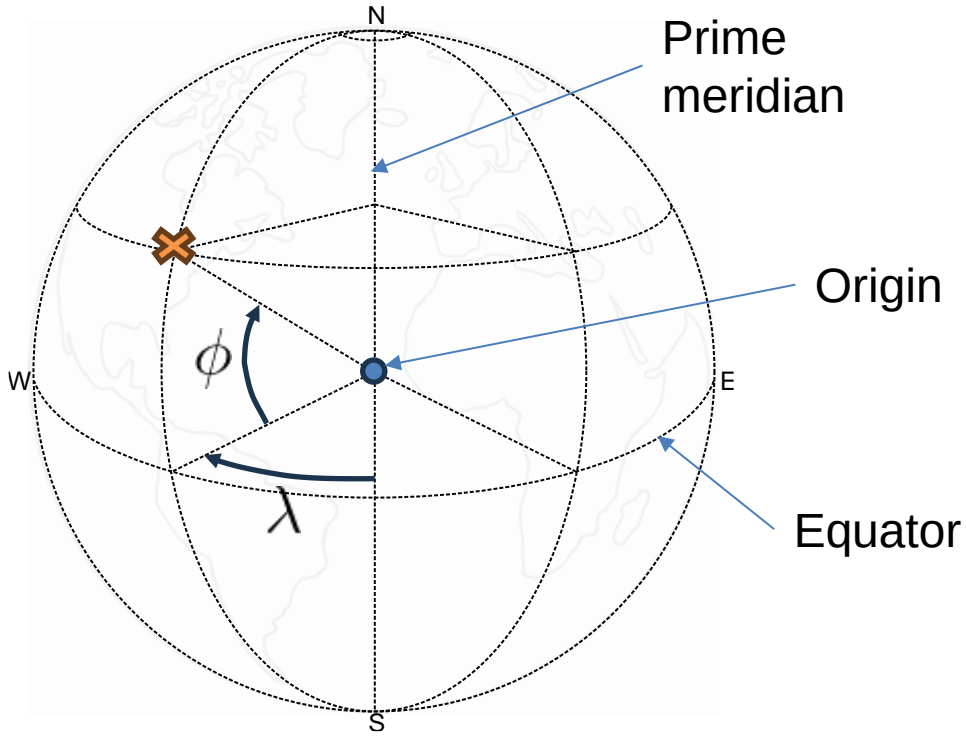
Position relative to camera



Pixel coordinate in image

Inertial Frame

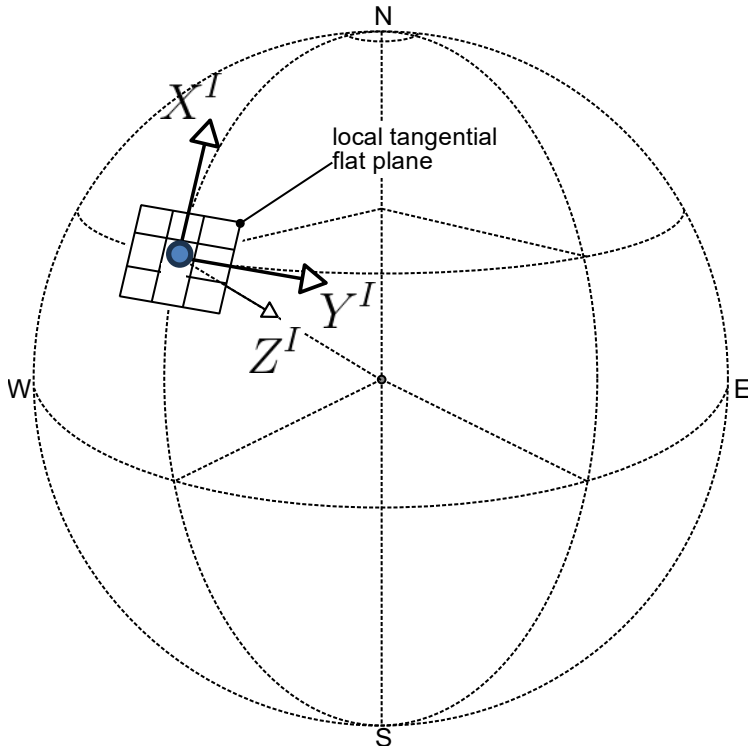
- Inertial frame defines position relative to the planet
- Typically expressed as latitude, longitude specifying point on planet surface
- e.g. 53.469356 N, -2.234126 E
- Symbols are:
  - Latitude: Phi ϕ
  - Longitude: Lambda λ



Origin	Centre of Earth
Orientation/ System	Spherical, positive North/East
Scale/Units	Degrees

Earth-fixed Frame

- Mobile robotics algorithm design and implementation is assumed to occur within a small region with its own local inertial frame.
- Define local flat plane around centre of expected motion
- e.g. (10,20,-35)
- Symbols are:
  - Northing location: X^I
  - Easting location: Y^I
  - Altitude: Z^I

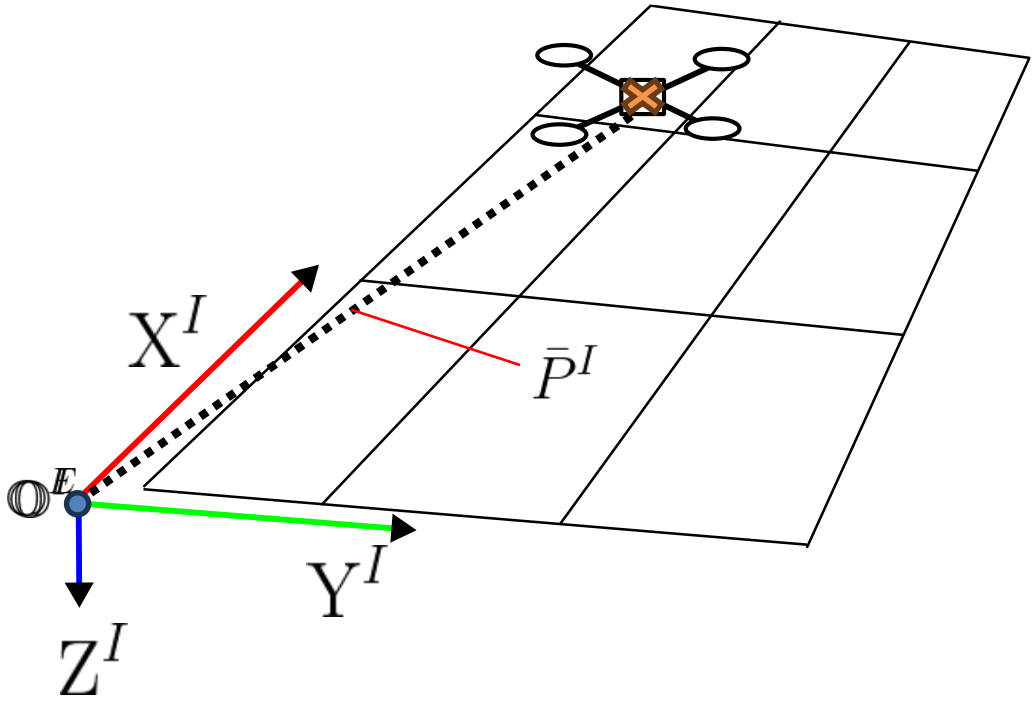


Origin	Point on Earth surface
Orientation/ System	Cartesian Positive – North, East, Down (NED)*
Scale/Units	Meters

Earth-fixed Frame (local inertial)

- A robot will have a position and orientation
- Often expressed within the inertial frame as two transforms
- Combined position and orientation known as ‘pose’
- Pose of body within Earth-fixed:
  - Position Earth: $\bar{P}^I = \begin{bmatrix} x^I \\ y^I \\ z^I \end{bmatrix}$
  - Orientation Earth $\Theta^E = \begin{Bmatrix} \phi^E \\ \theta^E \\ \psi^E \end{Bmatrix}$

Origin	Point on Earth surface
Orientation/ System	Cartesian Positive – North, East, Down (NED)*
Scale/Units	Meters

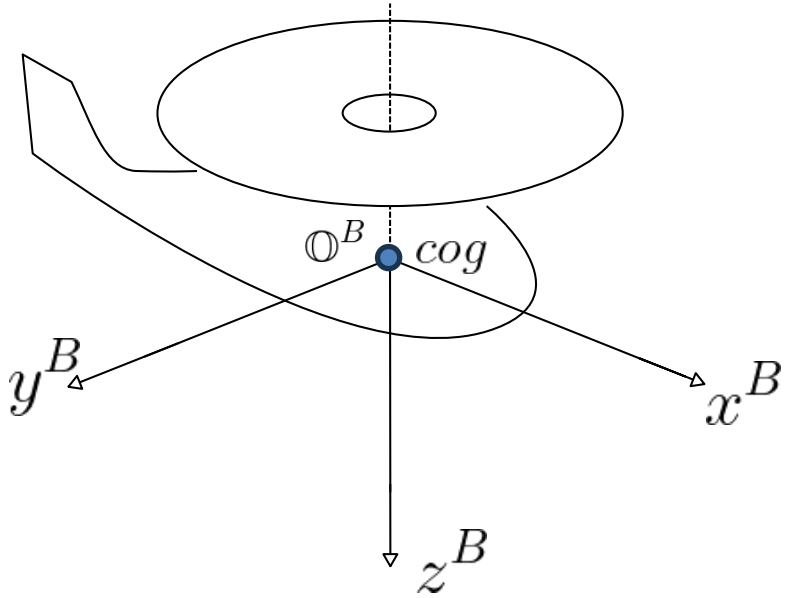


Body-fixed Frame

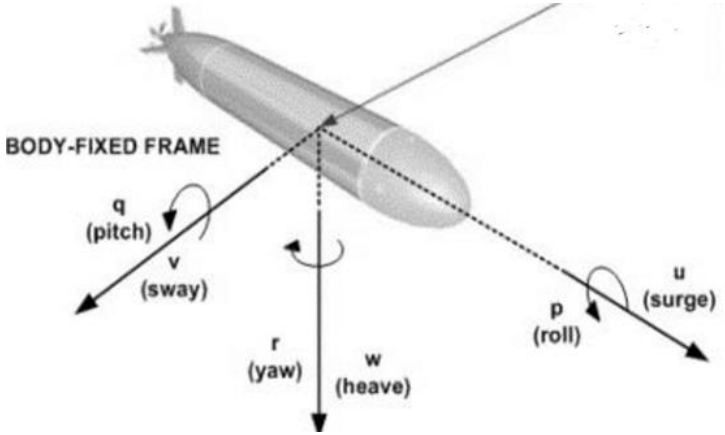
- The body frame of reference is fixed to the robot structure and moves with it
- Allows position and orientation of sub-components to be expressed
- Origin is often at the centre-of gravity (cog) (but not necessary)
- Axes aligned to distinguishing features (i.e. towards front/nose/primary axis)
- Position and orientation (pose) relative to body origin:

$$\bar{P}^B = \begin{bmatrix} x^B \\ y^B \\ z^B \end{bmatrix} \qquad \Theta_B = \begin{Bmatrix} \phi_B \\ \theta_B \\ \psi_B \end{Bmatrix}$$

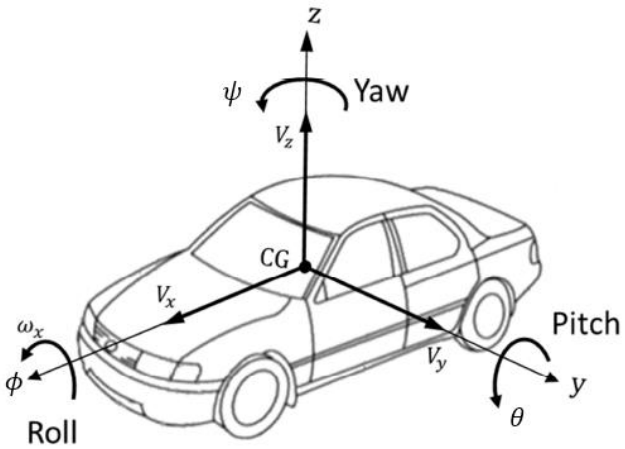
Origin	Centre of gravity
Orientation/ System	Cartesian Positive – Forwards, Starboard, Down
Scale/Units	Meters



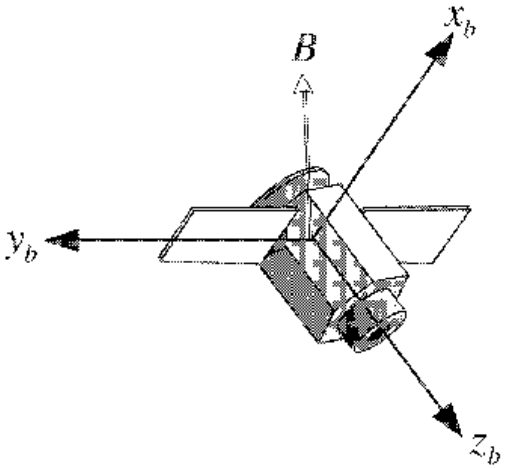
Body-fixed frame



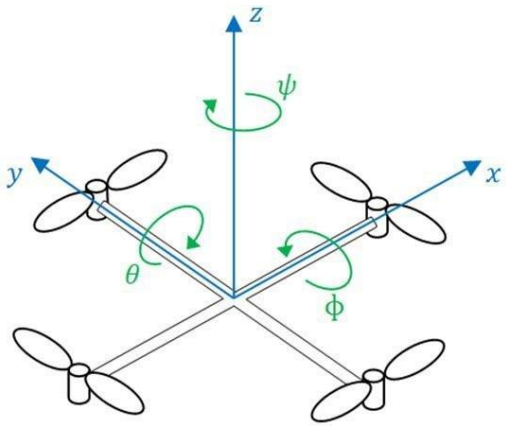
Underwater vehicles



Ground vehicles



Space vehicles



Aerial vehicles

Body-fixed frame

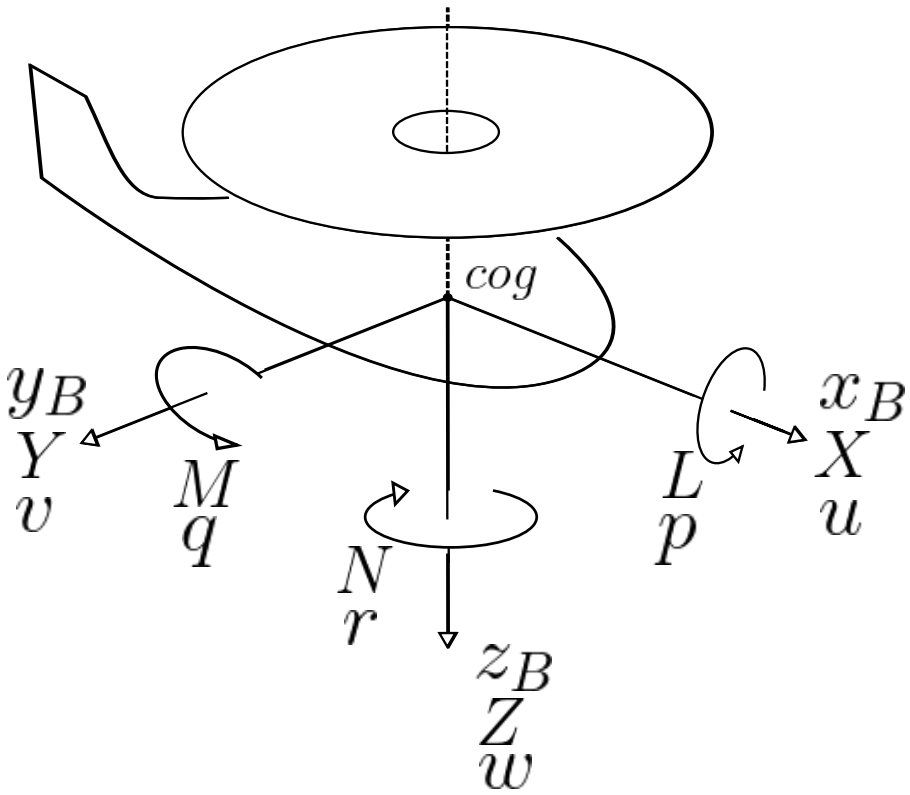
- Use body frame to define many motions and notation

Translational

Position vector $\bar{P}^B = \begin{bmatrix} x^B \\ y^B \\ z^B \end{bmatrix}$

Velocity vector $\bar{V}_B = \begin{bmatrix} u \\ v \\ w \end{bmatrix}$

Forces vector $\bar{F} = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$



Rotational

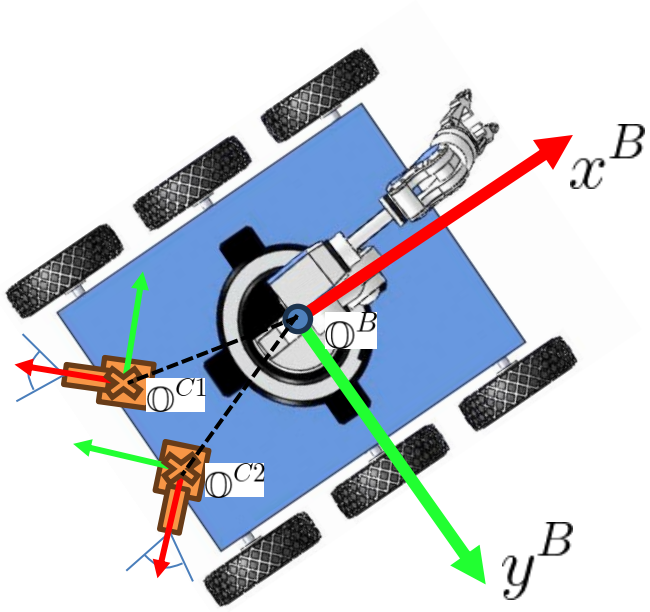
Orientation triplet $\Theta_B = \begin{Bmatrix} \phi_B \\ \theta_B \\ \psi_B \end{Bmatrix}$

Angular rate vector $\Omega = \begin{bmatrix} p \\ q \\ r \end{bmatrix}$

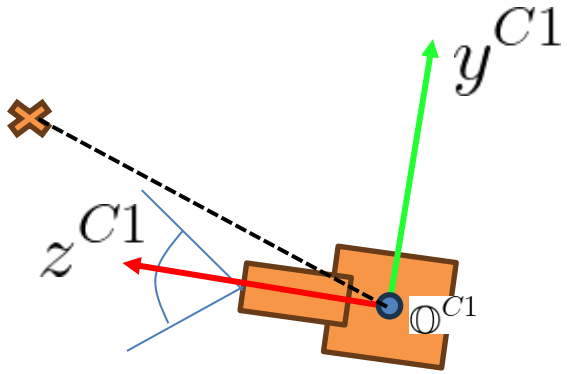
Moments vector $\bar{M} = \begin{bmatrix} L \\ M \\ N \end{bmatrix}$

Sensor-fixed frame

- Within the body-fixed frame there might be components or sub-systems of the robot which are not coincident or aligned to the body-fixed frame.
- Generally, each sub-frame has a pose relative to the parent frame



Cameras (sensors) have a pose relative to the robot body-fixed frame



Origin	Centre of focal plane
Orientation/ System	Cartesian Positive – left, down, forwards
Scale/Units	e.g. mm

Points within camera (sensor) sight have pose relative to the camera-fixed frame

Transformations

- Convert values from one frame of reference in another
  - “project into reference frame”
- When using Cartesian for all nested frames, this can be the same process applied in sequence.
  - Multiple successive transforms can be combined into one transform
- Need to introduce rotation notation and direction

$\mathbb{R}_{ab}$

Θ_{ab}

Q_{ab}

Subscript is order specific and used to indicate this is the rotation from reference frame a to reference frame b

Inertial to Earth-fixed transformation

- Subject of another lecture/tutorial/self-study

- The Inertial-to-Earth transform is given in appendix of:
  - **Wood, K.**, Thomas, H., Watson, M., Calway, A., Richardson, T., Stebel, K., Naismith, A., Berthoud, L., Lucas, J. *"Measurement of three dimensional volcanic plume properties using multiple ground based infrared cameras"*. ISPRS Journal of Photogrammetry and Remote Sensing 154. **2019**. https://doi.org/10.1016/j.isprsjprs.2019.06.002

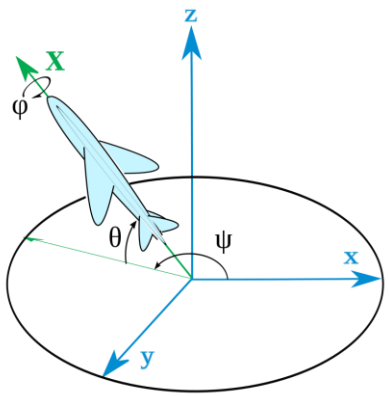
- The inverse Earth-fixed to inertial is more tricky!

Rotation representations

- There are several methods for representing orientation

- **Euler angles**
  - Useful because they are human interpretable
  - Common in aerospace
- **Direction cosine matrix** (rotation matrix)
  - Useful because they can be used directly in systems of equations
- **Quaternions**
  - Useful because they don’t suffer from singularities
  - Remain consistent at small or large angles
- **Axis-angle**
  - Useful rotating about an axis not centred on the origin

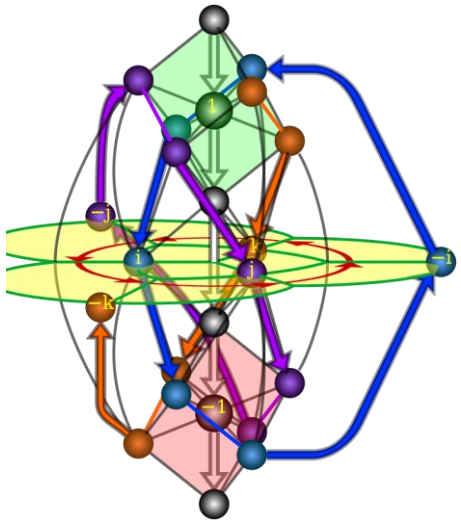
Euler angles



- A triplet of three angles ← not a vector!
- Orientation axis order must be known/shared
- Human readable (when expressed in degrees)
- Applied to 3D vectors by successive single axis rotations around (x, y, z) axes in any order
- Common order for aerial vehicles is: yaw, pitch, roll
  - but written roll (Phi), pitch (Theta), yaw (Psi)
- Difficult to combine multiple rotations
- Suffers from gimbal-lock

$$\Theta_B = \begin{Bmatrix} \phi_B \\ \theta_B \\ \psi_B \end{Bmatrix}$$

Quaternions



- 4 element array ← also not a vector!
- Quaternion norm must always have unit magnitude to represent a rotation

- Related to axis-angle, comprised of:

s – scalar roughly represents the axis of rotation

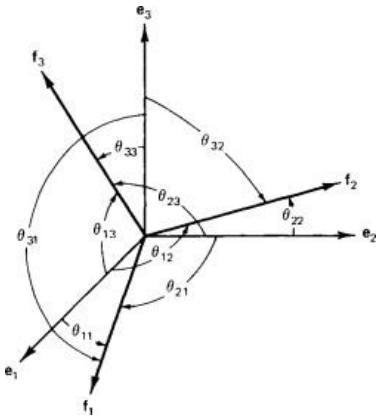
$\bar{v}$ – vector roughly represents the angle of rotation

$$\mathbf{Q} = \begin{bmatrix} q_w \\ q_x \\ q_y \\ q_z \end{bmatrix} = \begin{pmatrix} s \\ \bar{v} \end{pmatrix} = \begin{pmatrix} cos\left(\frac{\theta}{2}\right) \\ \bar{u} sin\left(\frac{\theta}{2}\right) \end{pmatrix}$$

$$\|q_1 \quad q_2 \quad q_3 \quad q_4\|_2 = 1$$

- Application via quaternion mathematics
- Allows smooth transitions through gimbal-lock

Direction Cosine Matrix (DCM)



- A 3x3 matrix representation
- Can be used directly within linear vector/matrix algebra
- Matrix only valid as rotation when orthogonal
  - determinant must be exactly +1
- Simple to combine multiple successive rotations
- Not human readable
- Can be constructed from a series of 1D axis rotations

$$\mathbb{R}_{ab} = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix}$$

Applying a rotation DCM

- To rotate 3 element vector $\bar{a}$ by DCM $\mathbb{R}_{ab}$ to form vector $\bar{b}$, use vector matrix multiplication.

$$\bar{b} = \mathbb{R}_{ab}\bar{a}$$

- …that is all
- Very convenient with programs such as Matlab or libraries such as Python numpy which perform matrix/vector math
- The inverse is also simple:

$$\bar{a} = \mathbb{R}_{ab}^{\mathrm{T}}\bar{b}$$

because $\mathbb{R}_{ab}^{\mathrm{T}} \equiv \mathbb{R}_{ab}^{-1}$ For orthogonal matrices

Applying a rotation Euler angles

- To rotate 3 element vector $\bar{a}$ by Euler angle triplet Θ_{ab} to form vector $\bar{b}$, first construct a DCM[1]…

$$\mathbb{R}_{ab} = f(\Theta_{ab}) = \begin{bmatrix} c\theta c\psi & c\theta s\psi & -s\theta \\ (-c\phi s\psi + s\phi s\theta c\psi) & (c\phi c\psi + s\phi s\theta s\psi) & s\phi c\theta \\ (s\phi s\psi + c\phi s\theta c\psi) & (-s\phi c\psi + c\phi s\theta s\psi) & c\phi c\theta \end{bmatrix}$$

- …where $s(\cdot)$ and $c(\cdot)$ are shorthand notation for sine and cosine functions.
- Then rotate as with DCM

$$\bar{b} = \mathbb{R}_{ab}\bar{a}$$

Remember definitions

$$\Theta_B = \begin{Bmatrix} \phi_B \\ \theta_B \\ \psi_B \end{Bmatrix}$$

[1] - a derivation of this formula is given in [“Small Unmanned Aircraft: Theory and Practice” Beard & McLain. Chapter 2]

Applying a rotation Quaternions

- To rotate 3 element vector $\bar{a}$ by Quaternion Q_{ab} to form vector $\bar{b}$ requires quaternion multiplication.
- Construct augmented matrices then multiply as normal

$$\begin{bmatrix} 0 \\ b_x \\ b_y \\ b_z \end{bmatrix} = \begin{bmatrix} q_w & q_x & q_y & q_z \\ -q_x & q_w & q_z & -q_y \\ -q_y & -q_z & q_w & q_x \\ -q_z & q_y & -q_x & q_w \end{bmatrix} \begin{bmatrix} 0 & -a_x & -a_y & -a_z \\ a_x & 0 & -a_z & a_y \\ a_y & a_z & 0 & -a_x \\ a_z & -a_y & a_x & 0 \end{bmatrix} \begin{bmatrix} q_w \\ q_x \\ q_y \\ q_z \end{bmatrix}$$

- Discard 4th element of resulting 4 element vector

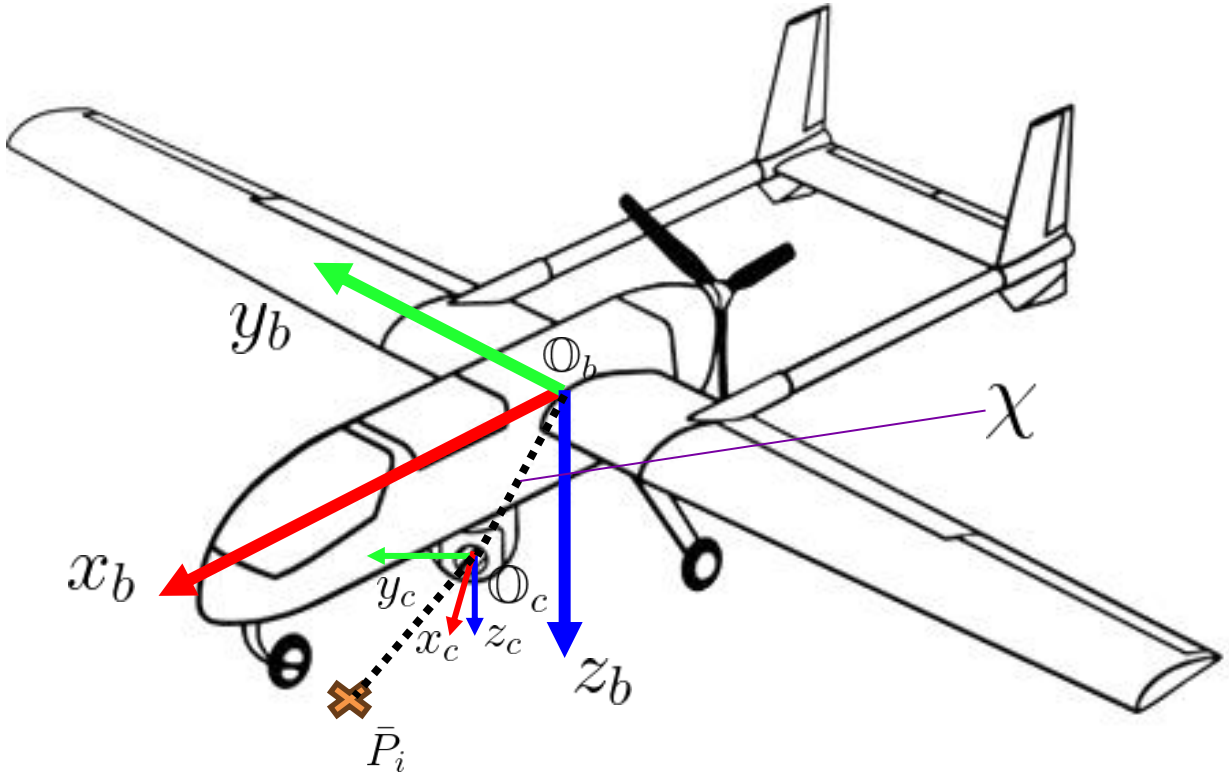
Remember definitions

$$Q = \begin{bmatrix} q_w \\ q_x \\ q_y \\ q_z \end{bmatrix}$$

More information on Quaternions and their application to rotations is given in "Small Unmanned Aircraft: Theory and Practice" Beard & McLain. Appendix B

Applying translation and orientation

- Assume we know the location of a point in a sensor frame of reference, and want to express in parent frame of reference (e.g. body frame).
- Assume sensor is both translationally and rotationally offset from body frame.



Point in camera frame $\bar{P}_i = \begin{bmatrix} x_i \\ y_i \\ z_i \end{bmatrix}$

Pose of camera frame relative to body frame $\chi = \begin{bmatrix} \bar{P} \\ \mathbb{R} \end{bmatrix}$

Translational position of camera frame $\bar{P} = \begin{bmatrix} x_{cam} \\ y_{cam} \\ z_{cam} \end{bmatrix}$

Relative orientation of camera frame $\mathbb{R}_{bc} = \begin{Bmatrix} \phi_{cam} \\ \theta_{cam} \\ \psi_{cam} \end{Bmatrix}$

Applying orientation and translation

- First rotate point by camera orientation to get intermediate point,

$$\bar{P}_i' = \begin{bmatrix} x_i' \\ y_i' \\ z_i' \end{bmatrix} = \mathbb{R}_{bc}\bar{P}_i^c \qquad ①$$

- Then add translational offset,

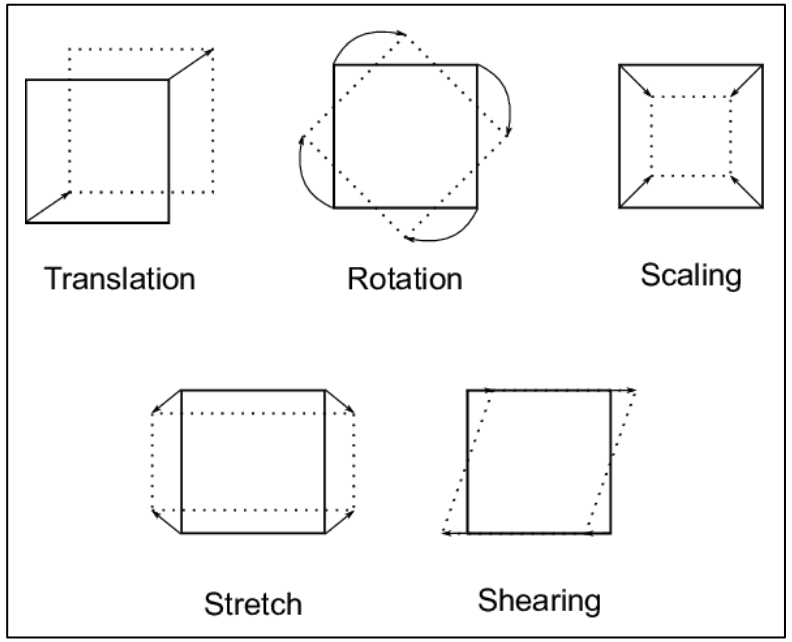
$$\bar{P}_i^b = \bar{P}_i' + P_{cam}^b \qquad ②$$

- This can be written as a single equation,

$$\bar{P}_i^b = \mathbb{R}_{bc}\bar{P}_i^c + P_{cam}^b \qquad ① \ \& \ ②$$

Affine transform

- The full affine transformation allows for simultaneous transformation of linear reference frames with,
  - Rotation
  - Skew/Shear
  - Mirroring
  - Stretching/scaling
  - Translation/offset
- …in multiple axes
- We are only interested in
  - Translation and rotation



The general affine operations expressed in 2D

Affine transform (homogeneous transform)

- To transform 3 element vector $\bar{a}$ by DCM $\mathbb{R}_{ab}$ and translation $\bar{P}^a$ to form vector $\bar{b}$, construct an augmented matrix…

$$\bar{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} c_{11} & c_{11} & c_{11} & x^a \\ c_{11} & c_{11} & c_{11} & y^a \\ c_{11} & c_{11} & c_{11} & z^a \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\text{affine transformation matrix } M_{ab}} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ 1 \end{bmatrix}$$

- …then multiply using normal vector/matrix mathematics
- Discard 4th element of resulting 4 element vector
- The 4x4 homogeneous transformation matrix can be composed of two or more affine transformations

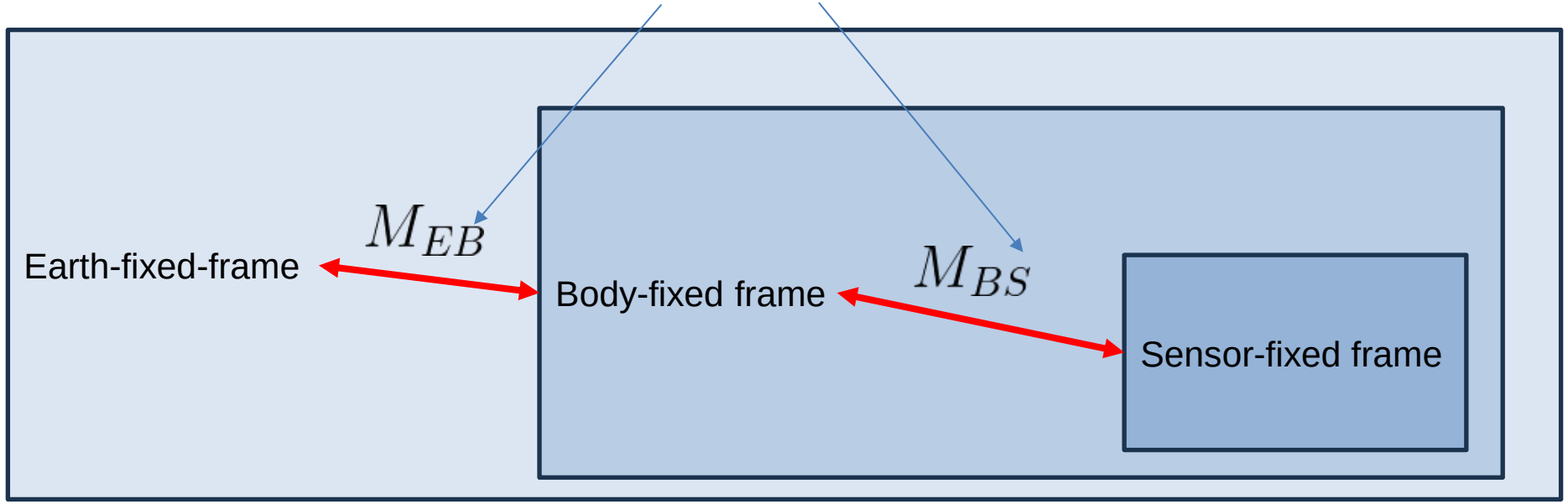
Remember definitions

$$\bar{P}^a = \begin{bmatrix} x^a \\ y^a \\ z^a \end{bmatrix}$$

$$\mathbb{R}_{ab} = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix}$$

Multiple affine transforms

Transformations defined by pose (position and orientation) relative to parent frame, expressed as a single affine transformation matrix.



$$\bar{P}_i^E = \underbrace{M_{EB} \cdot M_{BS}} \cdot \bar{P}_i^S$$

$$\bar{P}_i^E = M_{ES} \cdot \bar{P}_i^S$$

Multiple affine transformations multiplied together result in another valid transformation.

Review

- Definitions of reference frames and coordinate systesm
- Nested reference frames
- Pose – position and orientation
- Rotation representations
- Applying rotations
- Combined rotation and translation